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The Role of Zoos and Aquariums in a Changing World

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Abstract

Zoos and aquariums have evolved significantly. From their origins as enclosures for the mere entertainment of the public, these institutions have undertaken new functions responding to the biodiversity crisis and social demands. Modern zoos and aquariums have the opportunity to educate people, contribute to species conservation, and produce animal-related research. However, there is increasing criticism toward the outcomes of their actions and the holding of species in their facilities. This review offers an integrated analysis of the state of knowledge about the role that zoos and aquariums play today. It describes their performance regarding their conservation, education, and research functions, highlighting general patterns and offering future perspectives. It identifies some challenges common to all these institutions, concluding that the way they keep up with the ever-growing social and environmental expectations will be decisive hereafter.

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1. INTRODUCTION

The concept of zoos and aquariums (hereinafter zoos) has evolved significantly. Whereas in the last century zoos were understood as places of entertainment designed for the simple exhibition of specimens, nowadays these enclosures are much more complex, offering a space with defined learning, awareness, research, and conservation functions (1, 2). Given the high number of people visiting zoos each year, these institutions have an enormous opportunity to educate people (3, 4) and help biodiversity conservation. However, there is increasing criticism toward the holding of species in their facilities, the outcomes of their conservation actions (especially when it comes to breeding and reintroduction), and their lack of commitment to in situ conservation, among other factors (5, 6).

We analyze the state of knowledge of the role that zoos and aquariums play nowadays, from an integrated approach. Our intention is not to solve the current problems or offer definitive guidelines for these institutions. We aim to describe the directions modern zoos and aquariums are taking and how those may or may not help them in their ultimate goal of helping humanity and the biodiversity we are part of.

We begin with a brief history of zoos, showing the main milestones in the evolution of the current zoo concept (**Figure 1**). In the following four sections, we look into the functions that modern zoos have taken on in recent years. In the last section, we integrate that information and focus on future perspectives.

2. A HISTORY OF ZOOS AND AQUARIUMS

Although the beginning of zoos is tough to determine, the existence of menageries, as a collection of animals for display, can be reported in Egyptian, Roman, and Chinese civilizations more than 4,500 years ago (7). In these ancient times, zoos were a symbol of power and prosperity with the aim of entertaining people. Kings collected animals in menageries for their enjoyment and to demonstrate their power and sovereignty (8).

Zoos throughout history have adopted different roles. The most traditional version of zoos, as sets of enclosures exhibiting animals, originated in the thirteenth century. The world's first zoological park was established in Vienna in 1751. This menagerie, the Tiergarten Schönbrunn, is the oldest continuously operating zoo globally, but was not initially open to the general public. Today, the Ménagerie du Jardin des Plantes is recognized as the first truly public zoo (open in 1794) and the second oldest continuously operating one (8). In the nineteenth century, Sir Stamford Raffles founded the Zoological Society of London in 1826. Two years later, the London Zoo followed, which also became home to the world's first aquarium (9). By the 1990s, these collections aroused great interest in the public, and there were more than 10,000 zoos around the world (10).

These traditional zoos worked principally as showcases where animals were kept in cages for public entertainment. Zoos represented tourist attractions in urbanized areas, where visitor entrance fees financially supported zoo activities and collections (11). Some of these zoos were founded not only as collections of animals but also as places where scientists of the time studied these exotic species (12). However, in the early twentieth century, zoos conceived a new way of exhibiting animals, showing them in naturalized places. In fact, the Zoological Society of London turned an old, abandoned farm into what is now the Whipsnade Zoo in 1931, representing the first outdoor zoo in Europe (13).

From the 1960s onward, voices warning about the environmental crisis gained momentum, and zoos revisited their roles and started to take on conservation actions. As a consequence, many institutions started captive breeding programs as an ex situ option to reintroduce endangered animals into their natural habitat, thus contributing to species survival (12, 14). At first, modern

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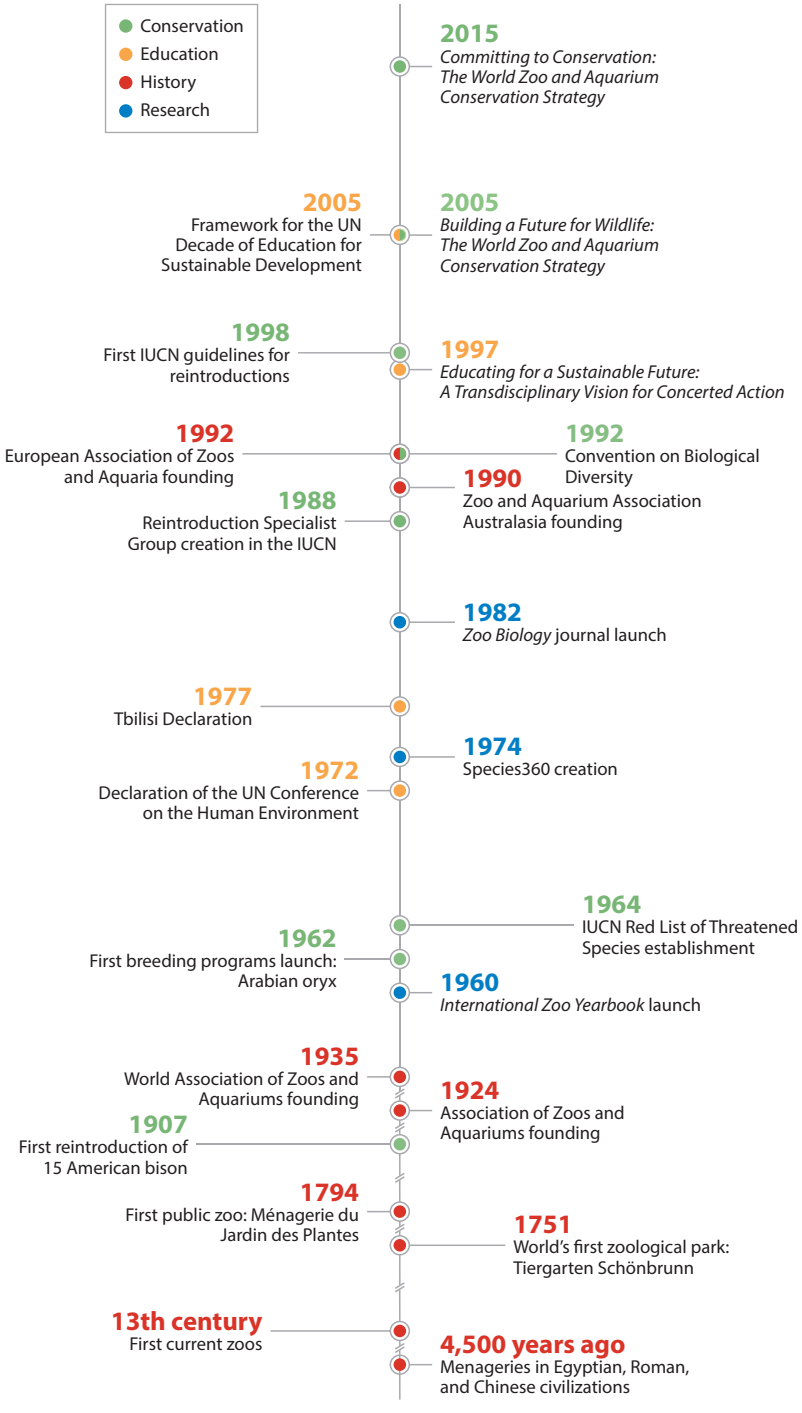

Figure 1

Timeline of principal milestones in the history of zoos and aquariums. Abbreviation: IUCN, International Union for Conservation of Nature; UN, United Nations.

zoos were considered places similar to Noah's ark, the conservation roles of which were limited to breeding and reintroduction programs (14, 15). The development of international initiatives such as the Convention on Biological Diversity (launched in 1992) spurred the advancement of a new concept of zoos and aquariums as instruments with a social and especially conservationist function (10, 16, 17) beyond ex situ conservation initiatives.

In this way, the role of modern zoos is not only entertainment but also conservation, education, and research. Contemporary zoos must ensure that people are aware of this role beyond the mere function of entertainment. However, the image of zoos tends to focus more on the entertainment they offer than on showing their role in education, conservation, and research (7).

In recent years, the social perception and objectives of zoos have evolved considerably, entailing legal changes. International and local directives establish the guidelines and obligations to ensure animal welfare, contribute to ex situ conservation of threatened species, and develop specific educational programs to raise awareness of biodiversity conservation (18).

Global initiatives and conventions represent strong support for conservation and research in zoos. However, this has not materialized in a notable development of these functions (19–21), due to several factors. Few zoos have the facilities or staff necessary to conduct research and offer valuable tools for conservation. Finally, although administrations promote conservation and research, there is no incentive for zoos to conduct conservation-relevant research (14, 18).

Zoos and aquariums face a great deal of criticism due to rising concerns about their respect for animal welfare (22, 23). Some organizations and movements claim that there are no reasons to maintain an animal in captivity and argue that zoos and aquariums should be closed down. These organizations believe that animals suffer in captivity, even in naturalized environments, because they cannot behave as they would in the wild. Furthermore, for these organizations, zoos' educational function is not significant enough compared to the entertainment they offer, implicitly teaching people that keeping animals from their natural environment is acceptable (6).

3. CONSERVATION OF BIODIVERSITY

One of the most relevant goals of modern zoos is to cooperate on biodiversity conservation (1, 20). However, a significant number of zoos and aquariums lack this function among their top priorities (1, 24). Most current zoo research is concerned with animal biology and welfare, sometimes with little use for conservation (18, 25). In addition, as mentioned above, the work of zoos in conservation is increasingly criticized at present (23, 26). This section reviews conservation actions and programs in zoological institutions and their relevance to biodiversity conservation.

How can modern zoos help conservation? There are two approaches in this regard: (*a*) ex situ, which entails actions out of the natural habitat of the species (e.g., captive breeding programs), and (*b*) in situ, which refers to actions in the species' habitat (e.g., animal reintroductions). Traditionally, these institutions have centered their conservation efforts on the reintroduction of specimens into their natural habitats, focusing on ex situ conservation activities, and maintaining genetic diversity and healthy captive populations (18, 27, 28). When tackling Extinct-in-the-Wild (EW) species, zoos might be crucial in maintaining viable captive populations through ex situ programs, acting as Noah's arks while waiting for a better future in which environmental conditions allow returning these species to their native range (29). Zoos also carry out relevant educational and research tasks concerning their conservation activities (3, 27), which contribute to conservation indirectly (20, 28).

It has been argued that conservation should focus primarily on habitat protection. The Strategic Plan for Biodiversity 2030 includes as one of its commitments to ensure the conservation trends of habitats and species by 2030 (30). However, keeping animals in zoos should be taken as a complementary action, especially for amphibians whose decline has been caused by chytridiomycosis

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(15, 31, 32). The Convention on Biological Diversity, in its ninth article, recommends the adoption of ex situ measures and the establishment of facilities for ex situ conservation, preferably in species' country of origin (33, 34).

For these reasons, it has become necessary to consider an integrated approach to conservation that includes ex situ and in situ strategies as complementary to adequate management strategies (16, 28, 33). The International Union for Conservation of Nature (IUCN) Conservation Planning Specialist Group proposed the so-called One Plan Approach (35), encouraging collaborative work toward conservation among all responsible stakeholders in ex situ and in situ conservation actions (28, 36).

Conserving biodiversity is a complex goal that requires the contribution of many stakeholders, and zoos have cooperated with government institutions, nongovernmental organizations, researchers, and other actors in several projects (37). The recovery of the Island fox (*Urocyon littoralis*) from the Channel Islands is a good example. This canid was downlisted from Critically Endangered in 2004 to Near Threatened in 2013 after a recovery program integrated by different agents (38). Zoological institutions contributed by developing guidelines for husbandry and genetic diversity management, offering veterinary care, and launching conservation education programs for this species (38).

If zoos totally aligned to this function of conservation, we would expect them to principally house threatened species. But in reality, zoos are taxonomically biased in favor of mammals and birds (39–42) over reptiles, amphibians, fish, and invertebrates (43), and they also house non-threatened species in higher numbers than threatened ones (42). In addition, the captive breeding programs of the principal associations of zoos and aquariums do not seem to focus on threatened species (**Figure 2**). Zoos house and exhibit principally charismatic species that help fulfill their entertainment role and generate enough income to keep operating (17, 44). At this crossroad between housing threatened species and charismatic ones, zoos must decide whether to follow the Noah ark's path or consider strategic planning to include other species to contribute to education, fundraising, and habitat protection actions that might help species in the wild (17).

To act as conservation centers, modern zoos should focus their efforts on breeding, education, and research for wildlife conservation (1, 14, 45). Moreover, they must balance their ex situ and in situ activities, increase their financial support for in situ activities, include conservation scientists in their work, and communicate their conservation work to visitors (28, 46).

3.1. Ex Situ Conservation Projects Promoted by Zoos and Aquariums

According to Target 4 of the last draft of the Post-2020 Global Biodiversity Framework, ex situ conservation should be included as a management action for species recovery and preservation (30). Some ex situ activities of zoos include population management, rehabilitation, public education, research, and husbandry (47, 48).

The most commonly known ex situ conservation activity is captive breeding. One of the reasons to develop this type of program is to have an insurance population in case a catastrophe occurs in the wild (45). Another reason might be protecting animals at their higher-risk stage of life, such as the eggs of some threatened turtles (49).

Zoological institutions cooperate through these breeding programs to maintain sustainable populations of species, which improves the results of the reintroduction of captive-bred species (20). Ex situ conservation programs are among the most frequent actions to prevent the extinction of mammals and birds (32, 37). Breeding programs have played an invaluable role in recovering at least 17 threatened species (47). For example, the Asian crested ibis (*Nipponia nippon*), Iberian lynx (*Lynx pardinus*), and red wolf (*Canis rufus*) would have become extinct without captive reinforcements and translocations (37).

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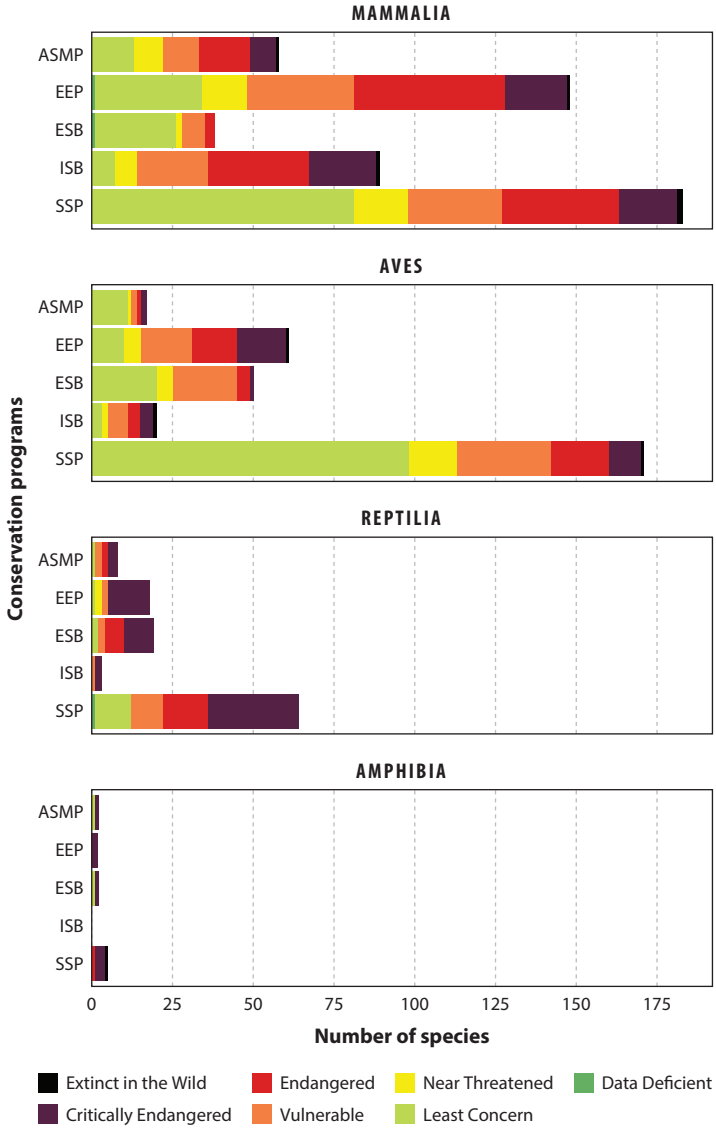

Figure 2

Captive breeding programs of principal associations of zoos and aquariums, distinguished by class and risk of extinction according to the International Union for Conservation of Nature Red List. Abbreviations: ASMP, Australasian Species Management Program (Zoo and Aquarium Association); EEP, European Endangered Species Program; ESB, European Studbook (European Association of Zoos and Aquaria); ISB, International Studbooks (World Association of Zoos and Aquariums); SSP, American Species Survival Plan (Association of Zoos and Aquariums).

Zoos and aquariums usually manage breeding programs in affiliation with other institutions around the world. Some zoos have played a major role in programs created especially for threatened species, such as the Golden Lion Tamarin Conservation Program (20, 50). Other successful recoveries of species include those of the black-footed ferret (*Mustela nigripes*), California condor

(*Gymnogyps californianus*), European bison (*Bison bonasus*), whooping crane (*Grus americana*), and Mauritius parakeet (*Psittacula eques*) (20, 47, 51, 52).

There have always been concerns related to captive breeding: hybridization risk, high cost, problems of moving animals among zoos, enough space for all animals in need, or no habitat for reintroduction (32, 51, 53). Ex situ conservation is not easy and is very expensive. The estimated cost of breeding a single animal for reintroduction reported by Perth Zoo was $6,546 in 2000, and food expenses for mammals in Chester Zoo exceed $700,000 per year (54). For some species, protecting their wild habitat is cheaper than maintaining them in captivity (20). Generally, in situ protection is more cost effective than captive breeding (55). However, changes happen in the wild, so captive breeding programs must remain an option (36, 56).

According to some studies, zoos should assign at least 10% of their income to biodiversity conservation (57); however, a study from the Association of Zoos and Aquariums (AZA) showed an average expenditure of only 0.1% (54). Also, zoos invest in conservation unevenly. Several zoos accomplish by far their conservation duties, whereas others invest nothing—or just a little—in this aspect. For example, Perth Zoo reported in 1999–2000 that the cost of breeding seven threatened species for reintroduction was $1,066,951, whereas their income in that period was $5,909,138. This expenditure represented 18% of their revenue (20).

The fact that a species is included in a population management program does not guarantee its protection or an improvement in its conservation status. The Bornean orangutan (*Pongo pygmaeus*) is subject to ex situ conservation management and included in the European Endangered Species Program. However, its wild population has decreased by 25% in the past 10 years (58), and the IUCN Red List catalogs the species as Critically Endangered. Even so, although reintroductions sometimes do not show successful results, and they present some difficulties and disappointments (50, 51), all in all this is today (and very likely in the future) an effective tool for biodiversity conservation (50, 59, 60).

3.2. Maintenance of Species Extinct in Nature

In 2003, the IUCN listed 30 species of animals as EW (61), meaning that they exist only in captive collections. Currently, this number has increased to 37, with amphibians, partulid snails, and some actinopterygians being most represented (62). Some of the actions taken to preserve these EW species are the conservation of their native refuges, maintenance in captive facilities (private and zoos), and captive breeding programs (29, 62).

Zoos have played an essential role in the successful recovery of EW species. This is the case of the Guam rail (*Hypotaenidia owstoni*), Przewalski's horse (*Equus ferus przewalskii*) (37), California condor (*G. californianus*), and black-footed ferret (*M. nigripes*), which have been reintroduced successfully (54, 63). Conversely, the Socorro dove (*Zenaida graysoni*), Guam kingfisher (*Todiramphus cinnamominus*), Alagoas curassow (*Mitu mitu*), and Père David's deer (*Elaphurus davidianus*) can be found only in ex situ facilities, as no viable populations can survive in their native habitats (29, 48). So, is it worth keeping EW species in zoos that can no longer live in their native places? Although at present, conditions and impacts in their native distribution area and population situation make it impossible to reintroduce these species, we do not know if their reintroduction will be possible and thrive in the future.

3.3. Contribution of Zoos and Aquariums to In Situ Conservation

In addition to ex situ conservation strategies, zoos also carry out in situ management plans (36, 64). More specifically, they support projects related to protection and habitat recovery, reintroductions, monitoring, and in situ captive breeding (10, 47, 48).

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Several conservationist and scientific institutions have significant in situ conservation projects in collaboration with zoos and aquariums, developed largely thanks to the zoos' incomes (25, 43, 65). Their actions comprise monitoring and managing habitat conservation, including the establishment and management of protected areas (1, 23, 54). Mainly, in situ conservation is developed in collaboration with local institutions and communities to facilitate and successfully achieve the project's goals (1, 20).

Concerning conservation priorities, zoos could help address the conservation of threatened local or regional species. The IUCN Species Survival Commission's (34) technical guidance for managing ex situ populations suggests that, as far as possible, ex situ management should occur within the range of and with similar climatic conditions to wild populations. In line with the One Plan Approach, all indications are that conservation projects are more likely to succeed when local stakeholders collaborate (1, 14, 20). For instance, Zoos Victoria in Australia identified 16 native Victorian species threatened with extinction by 2020 and decided to start conservation programs where appropriate (66).

Regarding economic contribution, in their annual report of 2020, the AZA reported a contribution of US$208.8 million to field conservation, with 942 species and subspecies benefiting from it (64). The European Association of Zoos and Aquaria (EAZA) contributed €17.8 million for conservation in the same year (67). According to their website, the Zoo and Aquarium Association Australasia participated in 629 conservation programs worldwide and funded AU$20.6 million for conservation programs in 2017. Finally, according to a World Association of Zoos and Aquariums survey, world zoos and aquariums spent approximately US$350 million on wildlife conservation in 2008, including ex and in situ actions on conservation (2, 68). Other studies suggest zoos and aquariums could contribute even more to in situ conservation with more funds, despite financial support being sometimes difficult to find (20, 34, 68).

Zoos can contribute in other ways besides providing funds, such as themed campaigns (15, 20, 69). For instance, the EAZA coordinates annual fundraising campaigns, raising more than US$300,000 per year for each themed campaign (54). Communication campaigns can also help biodiversity conservation through citizens' direct action. Wellington Zoo in New Zealand promoted a campaign that encouraged keeping cats indoors at night (preventing native wildlife predation), with a successful impact among visitors (69). Moreover, zoos can provide technical skills and capacity building to projects in the wild. Some examples are the Wildlife Conservation Society field veterinary program, the Chester Zoo's keeper outreach scheme, or the Training Centre at Jersey Zoo (54).

3.4. Reintroduction Programs as Flagship In Situ Conservation Actions

The release of animals into the wild as an in situ conservation activity dates back to the early twentieth century. The first reintroduction was probably the 1907 release of 15 American bison (*Bison bison*) into a reserve in Oklahoma (50, 60). Later, during the 1970s and 1980s, some mammal and bird reintroductions occurred, including those of the Arabian oryx (*Oryx leucoryx*) in Oman, golden lion tamarin (*Leontopithecus rosalia*) in Brazil, and peregrine falcon (*Falco peregrinus*) in North America (59, 60).

The first reintroduction programs lacked proper knowledge about the requirements for a successful reintroduction (59), as well as information on planning, resources, appropriate specimens, and post-implementation monitoring (60). To guide the development of reintroduction programs, the IUCN Species Survival Commission established the Reintroduction Specialist Group in 1988 (55, 60). In 1998, the IUCN published the first guidelines to help the design, implementation, and post-release monitoring of reintroduction processes (36, 53).

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The success of reintroduction projects has been largely analyzed in publications (53, 70). Beck et al. (71) found a success rate of 11% in 145 reintroduction projects with captive animals between 1900 and 1991. In a similar assessment of 180 case studies between 1980 and 2000, 26% were successful (36). However, these authors warned that these percentages must be interpreted carefully, as it is not easy to evaluate the success of a reintroduction project.

In this sense, these populations must be monitored to know their impact on the reintroduction of captive-bred species (20, 72). It is also essential to know if there is suitable habitat for the species and if the cause of decline has decreased (54). One example is the Arabian oryx (*O. leucoryx*) reintroduction programs, which were hampered by the illegal wildlife trade and the 1990 Gulf War (65, 73). In addition, it is better to use wild-caught animals than captive-bred ones (74). Captivity may diminish the ability to survive in the wild, cause poor health, or induce stress during the release process (60, 70, 73).

Despite these difficulties, reintroductions have increased over time (74–76). An analysis of reintroduction trends (including programs carried out by either zoos or other actors) from 1986 to 2017 revealed that the number of publications on animal release has increased, mammals and birds being the most common taxa released. Furthermore, of 213 analyzed species, 12.4% were Endangered and 8% were Critically Endangered, according to the IUCN Red List. Of all the reintroductions, 46% were successful, and 9.9% failed (76). Species whose conservation has improved due to reintroductions include the Grand Cayman blue iguana (*Cyclura lewisi*) and the Okarito kiwi (*Apteryx rowi*) (72, 77). Taxa for which reintroduction has been fundamental encompass mainly birds and mammals; these include even a change in their Red List status out of threatened categories, such as the greater stick-nest rat (*Leporillus conditor*) and the burrowing bettong (*Bettongia lesueur*) (57, 78).

To assess zoo involvement in reintroduction, *Global Reintroduction Perspectives* (*GRP*), published from 2008 to 2016, included an analysis of 243 projects worldwide, of which 35% had zoo or aquarium involvement and 20% provided captive-bred animals. Schwartz et al. (36) analyzed the information published in the *GRP* (2008, 2010, and 2011), finding that projects for mammals (72%) and fish (68%) had the highest success rates, followed by reptiles (56%) and birds (42%), with the lower levels (33%) for invertebrates and amphibians. Of all the studied projects, the ones directed at amphibians received the most support from zoos (36, 74). This research also found that EAZA member institutions were involved in 260 reintroduction projects, most of them for bird and mammal species (74). They reported their participation in reintroducing 32 species not held in their institutions. More than half of the species were European and native to the institution's country. Moreover, 53% of the species were threatened (74). Although in most cases zoos' principal contribution to reintroduction is the supply of animals, they also participate with funding, expertise, staff, equipment, and project coordination (71, 74, 75).

4. EDUCATIONAL ROLE

In the context of the global biodiversity crisis (79), zoos and aquariums may constitute valuable resources for environmental education and awareness because they reach a wide and varied audience (80–82). They offer citizens in increasingly urbanized societies an opportunity to experience nature in their immediate environment (6). However, their educational success depends on several elements, some related to the visitor (such as age, previous knowledge, or motivation for visiting the facility) (83), and others linked to the zoo or aquarium itself (such as exhibition design, educational resources, or how information is displayed) (3, 84).

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4.1. Resources for Education at Zoos and Aquariums

The main educational resources used in zoos and aquariums can be grouped into educational signs, lectures and guided visits, animal shows, exhibitions and immersive spaces, and information technologies. Educational signs offer written scientific information about each species, for instance, its common and scientific name, geographical distribution, behavior, and conservation status. Signs are the main educational resource and the one that adult visitors use the most (85, 86). Even so, few visitors read beyond the animal's name (83), either because they prefer to focus their attention on the specimen itself or because they deem the sign dissuasive due to its contents (too much text and technical language), design, or placement (87, 88). Clear and concise messages and an attractive design are essential for inviting visitors to use educational signs.

Compared to educational signs, lectures and guided visits entail direct communication between visitors and educators or curators. This simultaneously helps better transmit the educational message and entertains the visitor (83).

Other educational resources are animal shows. These displays are attractive to visitors and facilitate their connection with animals (89). However, there seems to be no agreement regarding their educational function: Whereas some authors consider shows a valuable tool for transmitting knowledge based on the visitor–animal connection (90, 91), others question their effectiveness in this regard (69). In addition, debate about the ethical implications of training animals for educational and recreational purposes is ongoing (12, 92). In this line, Fernandez et al. (93) highlight the need to find educational activities that do not stress animals, and Spooner et al. (5) suggest that future studies should compare the educational effectiveness of animal shows to that of spectacles that do not use actual animals.

Naturalistic exhibitions recreate animal habitats using actual vegetation, water, and sounds from nature. Immersive spaces, such as aviaries and butterfly houses, allow visitors to walk among animals who live in a recreated habitat. In both cases, animals suffer less stress than if they participate in animal shows, as they can easily hide from visitors. At the same time, this behavior can diminish the display's attractiveness for visitors who attend zoos looking for active and interactive animals (94, 95). From the educational point of view, exhibitions and immersive spaces allow visitors to see animals in their corresponding environment, thus offering a contextualized learning opportunity (46, 90, 96, 97). However, this educational potential drops when animals are hidden, because in that case visitors do not stop to observe the exhibition or read the educational signs (88, 98). According to Moss & Esson (91), the learning opportunities that a certain animal offers are directly proportional to the interest it awakens in visitors.

Finally, interactive screens, electronic games, tablets, and other digital assets are especially attractive for children and teenagers (87). These resources are controversial, as some authors advocate for keeping the zoo experience as real and "natural" as possible (3). The maintenance costs of these devices may also make them unaffordable for some zoos. In this regard, QR codes are a cheaper option to offer a significant amount of information in a small space (83). To summarize, naturalistic exhibitions and interactive materials' effect on visitors' knowledge and attitude are under discussion; whereas their didactic potential is widely accepted, the number of studies proving their efficacy remains insufficient.

4.2. Impact of Educational Resources and Programs

The educational potential of zoos goes beyond teaching about animals, as the lessons offered can transmit environmental messages to foster behavioral changes toward sustainability (81). From this point of view, these facilities have unique resources for education. For example, housing species currently threatened by climate change may help zoos teach the public about this phenomenon. In

this line, EW species hosted in zoos may provide a unique opportunity to teach about biodiversity loss.

All of the reviewed research papers report that zoos' educational resources, programs, and experiences effectively transmit new knowledge. Several describe how visitors retain such knowledge weeks or even months after their visit (69, 99–101). In contrast, there is some debate over the effects of zoos and aquariums on visitors' pro-environmental actions. Whereas some authors report that these facilities encourage such behaviors (100–102), others state that there is insufficient evidence to demonstrate that effect (99, 103, 104). In this regard, Moss et al. (105) point out that most studies deduce the positive educational impact of a zoo or aquarium experience from post-visit measures. Pretest–posttest designs are necessary to obtain more solid results. In addition, whether the effects on visitor attitudes and behavior last in the long term is unclear, because few studies consider long-term consequences (46, 69, 106, 107).

The best educational outcomes correspond to experiences that (*a*) demand active visitor participation (99), (*b*) involve animals (100), or (*c*) encourage visitors' emotional connection with animals (89, 102). Interviewed visitors and teachers consider it necessary for the zoo to offer informative materials after the visit (86, 108), such as a brochure describing daily actions to undertake in favor of conservation (81).

The main concern we found regarding zoos' educational impact relates to assessing visitors' actual change of behavior. Some authors raise concerns that published studies usually measure just the visitors' intention to modify their habits, but there is a poor association between intentions and behavior (109–111). Two main reasons likely explain this knowledge gap: the complexity of developing mid- or long-term studies in this matter and the impossibility of observing visitors' daily behavior to evaluate pro-environmental shifts. Swanagan (96) tried to overcome these hindrances by inviting visitors to sign a petition supporting an environmental cause. Such action does not mean a high degree of commitment but does help to validate to some degree the willingness to act. On this point, Nygren & Ojalammi (112) noted that the very nature of learning and behavioral change makes them difficult to measure because they are slow, complex, and interactive processes. In any case, this line of research must continue to overcome the methodological difficulties that such long-term studies entail.

4.3. Educational Challenges and Opportunities

To strengthen their educational function, zoos and aquariums must help visitors connect new information to their previous knowledge, interests, and experiences (12). This means that, in addition to passive resources such as signs, these facilities should tailor educational programs and strategies to specific groups.

One of the most common audiences of zoos and aquariums is school students. Some of the analyzed papers highlight the need to strengthen the collaboration between schoolteachers and zoo educators. A school visit to the zoo should be not an isolated event but an activity integrated into the course syllabus (46). To be effective, teachers and educators must carefully plan the visit; set goals; and work on them before, during, and after going to the facility (108). Several authors stress the importance of post-visit activities and educational materials to fix the ideas the zoo intended to transmit (86, 113) and to promote pro-conservation behaviors (107).

The difficulty of assessing learning and, especially, behavioral change highlights the need for longitudinal studies that go beyond measuring intentions and provide actual data about visitor changes in habits (110). In this line, Stoinski et al. (114) recommend zoos offer visitors on-site opportunities to turn conservation interests into action.

Learning is not just an intellectual activity but involves the whole person, especially when trying to promote pro-environmental attitudes. In this sense, Hughes (110) pointed out the need

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to assess the role of emotional attachment or connection in conservation learning. Some studies have been done in this line (115), but more research is necessary to reach precise conclusions.

Future studies in this area might address some gaps in knowledge. Firstly, Davey (98) points out that studies about zoo visitors happen mainly in Europe and America, whereas information about other countries is very scarce. Assessing the educational function of zoos and aquariums in places that welcome a significant number of tourists (such as certain African or South American countries) might be of great interest. Secondly, there is an opportunity to address gender differences in learning in these facilities. Gender may affect knowledge, attitude, and behavior outcomes (4, 99), but more studies are needed to draw solid conclusions in this regard. Finally, Learmonth et al. (107) suggest that future studies examine the differences in attitudes and beliefs between zoo visitors and the general public (non-visitors) to better assess these facilities' educational role.

5. RESEARCH CARRIED OUT BY ZOOS AND AQUARIUMS

Research is one of the core missions of modern zoos. Having access to a wide array of captive species offers an opportunity for researchers to gather some information that could be impossible to obtain otherwise. Much of this research is applied and oriented to problem-solving, and zoos have benefited from it, as they have been able to improve the well-being of the animals they house. However, zoos remain in the middle ground between in situ and laboratory research, so their results might not apply to real conservation cases in the wild (116).

5.1. More Than 60 Years Contributing to Research

Zoos have published their scientific production in both peer-reviewed and non-peer-reviewed formats since adopting the role of research institutes (116). The first scholarly publications date to the early 1960s, when the journal *International Zoo Yearbook* published its first issue. Since then, the number of peer-reviewed documents related to zoo research and authored by zoos has been growing steadily (25, 117–119). Not surprisingly, zoos have widely targeted zoo-oriented journals to publish their research. The top three journals compiling the scientific production of zoos globally and over time are the *Journal of Zoo and Wildlife Medicine*, *Zoo Biology*, and *International Zoo Yearbook* (25), all of them promoted by institutions related to zoos. However, we can also find zoo research in nonspecialized journals, suggesting zoos' desire to reach out to a broader public (19).

To date, more than 1,000 institutions belong to the network mobilized by Species360, the nonprofit nongovernmental organization that integrates global data about species housed in zoological institutions. Despite this high number of organizations potentially conducting research in their facilities, the pattern of contributions is highly skewed. Just 1% of these centers have published approximately 60% of scientific work coauthored by zoos, the majority of which has been taken over by North American institutions, followed by a few European institutions and one Asiatic institution (25).

So, what are the reasons for such uneven distribution of scientific production? The perceived factors limiting research conducted at zoos have remained the same since these institutions began including research in their agendas. In the 1990s, Finlay & Maple (120) launched a survey on North American members of the AZA and concluded that the lack of time, budget, and qualified research staff prevented zoos from researching. Loh et al. (119) found that the scientific productivity of AZA members was associated with an increase of $106 million in 10 years of the conservation and research budget. Of course, if more funding is allocated to research, the number of published papers will likely increase.

But not only resources influence this distribution. Staff conducting research at zoos perceive that successful scientific programs parallel support from the director, well-oriented research

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objectives, and human resources devoted to research (121). For example, zoos that included research as a mission statement had higher scientific production rates (119). Therefore, a board of directors committed to research is likely to increase the resources invested in it, by increasing research budgets, devoting more time for employee research, or even growing full-time research positions of qualified professionals.

When increasing qualified research positions in zoos is not possible, collaboration between zoos and academic institutions is an alternative to support zoo-based research. Zoos have a long collaboration history with other research institutions (27). Collaborative papers between zoos and other institutions rose 49% in 10 years in *Zoo Biology* (122, 123). Also, cross-zoo collaboration is necessary to get sufficient sample sizes to obtain robust results (124). However, whether collaboration between zoos and academic institutions fosters research is debatable, as no work has properly addressed this issue.

5.2. High Diversity of Topics, Low Diversity of Species

Researchers have characterized research output by zoos under different conceptual frameworks. Several have analyzed the contribution of a particular association (21, 118, 119, 125) or journal (122) or analyzed articles dealing with zoos' research (but not necessarily published by them) and articles published by zoos worldwide (25). The methods and temporal coverage of such works also vary, making comparing their results difficult. However, each gives a partial but valuable view of the landscape of research production in zoos, and some of the trends found individually are supported collectively.

Zoo research covers a wide variety of topics, such as husbandry, behavior, animal welfare, and nutrition. Kaufman et al. (116) provide a brief overview and examples of the issues covered by zoos. Escribano et al. (25) mapped research production by analyzing keyword frequency in articles authored by zoos worldwide, using them as a proxy for researched topics. Overall, the most popular keywords covered breeding, conservation, behavior, genetic diversity, and animal welfare topics. The study also allowed analysis of trends over time. Zoos' interests have changed over time, apparently influenced by other trends in general science (25). For example, breeding and related keywords, the most common topics addressed in the 1980s and 1990s (120, 123), gave way to keywords dealing with genetics and conservation when DNA techniques improved and the conservation crisis took center stage.

Unlike the topics (with breeding being prevalent), the species targeted for zoo research have not diversified over time. The literature widely agrees and supports that zoos' research is taxonomically biased. Overall, mammals dominate the landscape of zoo research globally (25, 40, 117, 120), despite the number of mammals housed in zoos being low compared to other taxa, such as birds or fish (40, 117). The 10 most-studied species are mammals and include one cetacean [common bottle-nosed dolphin (*Tursiops truncatus*)]; four carnivores [wolf (*Canis lupus*), domestic cat (*Felis silvestris catus*), lion (*Panthera leo*), and giant panda (*Ailuropoda melanoleuca*)]; two primates [chimpanzee (*Pan troglodytes*) and western gorilla (*Gorilla gorilla*)]; two proboscideans [African savanna elephant (*Loxodonta africana*) and Asian elephant (*Elephas maximus*)]; and one diprotodont [koala (*Phascolarctos cinereus*)] (25).

Another dimension is whether zoos orient their research toward threatened species (according to the IUCN Red List). Although the representation of threatened species in zoos is low (47), the research gives a slightly positive scenario, as species above Near Threatened status are mentioned in nearly half of all articles published by zoos (25). For example, 7 of the 10 aforementioned species fall within threatened categories (Critically Endangered: gorilla; Endangered: chimpanzee and Asian elephant; Vulnerable: lion, giant panda, African savanna elephant, and koala) (62).

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Mammals dominate research output by zoos for many potential reasons. Rose et al. (117) argue that researchers' interest in or access to such taxa could explain this trend. Melfi (40) points out that mammals are the most-studied species because we perceive they require higher animal welfare conditions. Finally, a species' threatened status apparently does not foster research about them as a whole, as the proportion of articles dealing with threatened species over time has stabilized (25).

6. FINAL REMARKS AND PERSPECTIVES

Biodiversity is being lost at unprecedented rates, and society is increasingly concerned about its future (79, 126). Year after year, global initiatives (e.g., Convention on Biological Diversity, Intergovernmental Science-Policy Platform on Biodiversity and Ecosystem Services, IUCN) have stressed the necessity of an integrative approach to curb this crisis (127–129). Despite ongoing heated debate about the morality of housing captive specimens, zoos have been called to build up their transformation process and take action (14).

Influenced by this demand, zoos have shifted their priorities to make room for education, conservation, and research activities in their agendas. These functions should drive their development plans and future strategies. Ideally, the achievement of these functions should be compatible with entertaining visitors and may not impact income and visitor numbers. Regardless of visitor motivation, most zoos depend on entrance fees to support their activities. Zoos will have to keep revisiting the way they operate, while offering entertainment and responding to their conservation and social obligations. On top of that, with an increasing sensitivity toward animals, zoos must always guarantee the welfare of the specimens they house: Animal welfare is a moral obligation that we cannot neglect (14, 93, 130).

Whether zoos are effectively fulfilling what current challenges and society demand is still debatable. One potential pitfall and source of criticism is the taxonomic bias found in the three functions. Although zoos house birds and fish more than other taxonomic groups (35), large-bodied mammals dominate zoo conservation and research actions. This taxonomic bias seems inconsistent with a proper approach to the three functions described. However, zoos have played a crucial role in the survival of several species by allocating significant resources to ex situ and in situ conservation programs. Besides, using charismatic animals as flagship species is a common strategy to increase funding and educate on conservation problems. This bias does not apply only to zoos. Conservation projects in general use these species to gain support from stakeholders and the general public (44, 131, 132).

Where zoos should place their priorities is yet to be decided. What is certain is that conservation efforts should align with global conservation priorities in collaboration with all responsible stakeholders (24, 52). The One Plan Approach seems promising to orient conservation actions that consequently will affect education and research in zoos. Several zoos have already adopted this approach, but more should follow, with improved global cooperation among institutions, to achieve their conservation goals successfully. However, zoos are mainly located in temperate regions (47), and the works related to them (research, conservation plans, perception studies) might reflect only Western points of view.

For all these reasons, the role of zoos and aquariums should be consistently revisited. However, current knowledge does not fully evaluate whether modern zoos are meeting the expected functional standards. Further evidence is needed regarding zoos' effectiveness as conservation, education, and research centers to demonstrate that the benefits outweigh the possible detriments to animal welfare and captivity (14). However, most work addressing the contribution of zoos across conservation, education, and research has been coauthored by people working in zoos and zoological associations. Although we do not question the rigorousness and reliability of these studies,

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an external and independent evaluation of their role would enrich the existing ones and also dispel doubts about possible conflicts of interest.

The future of modern zoos and aquariums will rely on their capacity to meet ongoing and ever-growing social and environmental expectations (14). Their plans should consider chiefly conservation priorities, which will inevitably be accompanied by education and research. Throughout this process, a profound change is necessary for many of these institutions. Some may disappear, others will have to adapt, and finally, a significant number of zoos will continue down the path they have already started. In this way, zoos may become an essential tool to meet the urgent needs that the biodiversity status demands.

DISCLOSURE STATEMENT

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LITERATURE CITED

1. Miller B, Conway W, Reading RP, Wemmer C, Wildt D, et al. 2004. Evaluating the conservation mission of zoos, aquariums, botanical gardens, and natural history museums. *Conserv. Biol.* 18(1):86–93
2. Barongi R, Fisken FA, Parker M, Gusset M. 2015. *Committing to Conservation: The World Zoo and Aquarium Conservation Strategy*. Gland, Switz.: World Assoc. Zoos Aquar. Exec. Off.
3. Wineman J, Piper C, Maple TL. 1996. Zoos in transition: enriching conservation education for a new generation. *Curator* 39(2):94–107
4. Collins C, Corkery I, McKeown S, McSweeney L, Flannery K, et al. 2020. Quantifying the long-term impact of zoological education: a study of learning in a zoo and an aquarium. *Environ. Educ. Res.* 26(7):1008–26
5. Spooner SL, Jensen EA, Tracey L, Marshall AR. 2019. Evaluating the impacts of theatre-based wildlife and conservation education at the zoo. *Environ. Educ. Res.* 25(8):1231–49
6. Whitehead M. 1995. Saying it with genes, species and habitats: biodiversity education and the role of zoos. *Biodivers. Conserv.* 4(6):664–70
7. Carr N, Cohen S. 2011. The public face of zoos: images of entertainment, education and conservation. *Anthrozoös* 24(2):175–89
8. Lindholm J. 2021. Zoo history. In *Zookeeping: An Introduction to the Science and Technology*, ed. MD Irwin, JB Stoner, AM Cobaugh, pp. 31–42. Chicago: Univ. Chicago Press
9. Rabb GB. 1994. The changing roles of zoological parks in conserving biological diversity. *Am. Zool.* 34(1):159–64
10. Matthews LH. 1976. The zoo: 150 years of research. *Nature* 261(5558):281–84
11. Mason P. 2000. Zoo tourism: the need for more research. *J. Sustain. Tour.* 8(4):333–39
12. Ballantyne R, Packer J, Hughes K, Dierking L. 2007. Conservation learning in wildlife tourism settings: lessons from research in zoos and aquariums. *Environ. Educ. Res.* 13(3):367–83
13. Kisling VN. 2001. *Zoo and Aquarium History: Ancient Animal Collections to Zoological Gardens*. Boca Raton, FL: CRC Press
14. Keulartz J. 2015. Captivity for conservation? Zoos at a crossroads. *J. Agric. Environ. Ethics* 28(2):335–51
15. Zippel K, Johnson K, Gagliardo R, Gibson R, McFadden M, et al. 2011. The Amphibian Ark: a global community for ex situ conservation of amphibians. *Herpetol. Conserv. Biol.* 6(3):340–52
16. Gippoliti S. 2012. *Ex situ* conservation programmes in European zoological gardens: Can we afford to lose them? *Biodivers. Conserv.* 21(6):1359–64
17. Fa JE, Gusset M, Flesness N, Conde DA. 2014. Zoos have yet to unveil their full conservation potential. *Anim. Conserv.* 17(2):97–100
18. Rees PA. 2005. Will the EC Zoos Directive increase the conservation value of zoo research? *Oryx* 39(2):128–31

Downloaded from www.annualreviews.org. Guest (guest) IP: 1.234.183.99 On: Mon, 03 Nov 2025 09:00:37

19. Ryder OA, Feistner ATC. 1995. Research in zoos: a growth area in conservation. *Biodivers. Conserv.* 4(6):671–77
20. Tribe A, Booth R. 2003. Assessing the role of zoos in wildlife conservation. *Hum. Dimens. Wildl.* 8(1):65–74
21. Pyott BE, Schulte-Hostedde AI. 2020. Peer-reviewed scientific contributions from Canadian zoos and aquariums. *Facets* 5(1):381–92
22. Olive A, Jansen K. 2017. The contribution of zoos and aquaria to Aichi Biodiversity Target 12: a case study of Canadian zoos. *Glob. Ecol. Conserv.* 10:103–13
23. Milstein T. 2009. "Somethin' tells me it's all happening at the zoo": discourse, power, and conservationism. *Environ. Commun.* 3(1):25–48
24. Conway W. 2003. The role of zoos in the 21st century. *Int. Zoo Yearb.* 38(1):7–13
25. Escribano N, Ariño AH, Pino-del-Carpio A, Galicia D, Miranda R. 2021. Global trends in research output by zoos and aquariums. *Conserv. Biol.* 35(6):1894–902
26. Hutchins M. 2006. Death at the zoo: the media, science, and reality. *Zoo Biol.* 25(2):101–15
27. Fernandez EJ, Timberlake W. 2008. Mutual benefits of research collaborations between zoos and academic institutions. *Zoo Biol.* 27(6):470–87
28. Mooney A, Conde DA, Healy K, Buckley YM. 2020. A system wide approach to managing zoo collections for visitor attendance and in situ conservation. *Nat. Commun.* 11:584
29. Trask A, Canessa S, Moehrenschlager A, Newland S, Medina S, Ewen J. 2020. Extinct-in-the-wild species' last stand. *Science* 369(6503):516
30. Nicholson E, Watermeyer KE, Rowland JA, Sato CF, Stevenson SL, et al. 2021. Scientific foundations for an ecosystem goal, milestones and indicators for the post-2020 global biodiversity framework. *Nat. Ecol. Evol.* 5(10):1338–49
31. Silla AJ, Byrne PG. 2019. The role of reproductive technologies in amphibian conservation breeding programs. *Annu. Rev. Anim. Biosci.* 7:499–519
32. Conde DA, Colchero F, Güneralp B, Gusset M, Skolnik B, et al. 2015. Opportunities and costs for preventing vertebrate extinctions. *Curr. Biol.* 25(6):R219–21
33. Gant JR, Mair L, McGowan PJK. 2020. Fragmented evidence for the contribution of ex situ management to species conservation indicates the need for better reporting. *Oryx* 55(4):573–80
34. Maunder M, Byers O. 2005. The IUCN Technical Guidelines on the Management of Ex Situ Populations for Conservation: reflecting major changes in the application of *ex situ* conservation. *Oryx* 39(1):95–98
35. Traylor-Holzer K, Leus K, Bauman K. 2019. Integrated Collection Assessment and Planning (ICAP) workshop: helping zoos move toward the One Plan Approach. *Zoo Biol.* 38(1):95–105
36. Schwartz KR, Parsons ECM, Rockwood L, Wood TC. 2017. Integrating *in-situ* and *ex-situ* data management processes for biodiversity conservation. *Front. Ecol. Evol.* 5:120
37. Bolam FC, Mair L, Angelico M, Brooks TM, Burgman M, et al. 2021. How many bird and mammal extinctions has recent conservation action prevented? *Conserv. Lett.* 14(1):e12762
38. Coonan TJ, Varsik A, Lynch C, Schwemm CA. 2010. Cooperative conservation: zoos and *in situ* captive breeding for endangered Island fox *Urocyon littoralis* ssp. *Int. Zoo Yearb.* 44(1):165–72
39. Brereton SR, Brereton JE. 2020. Sixty years of collection planning: What species do zoos and aquariums keep? *Int. Zoo Yearb.* 54(1):131–45
40. Melfi VA. 2009. There are big gaps in our knowledge, and thus approach, to zoo animal welfare: a case for evidence-based zoo animal management. *Zoo Biol.* 28(6):574–88
41. Ward PI, Mosberger N, Kistler C, Fischer O. 1998. The relationship between popularity and body size in zoo animals. *Conserv. Biol.* 12(6):1408–11
42. Martin TE, Lurbiecki H, Joy JB, Mooers AO. 2014. Mammal and bird species held in zoos are less endemic and less threatened than their close relatives not held in zoos. *Anim. Conserv.* 17(2):89–96
43. Godinez AM, Fernandez EJ. 2019. What is the zoo experience? How zoos impact a visitor's behaviors, perceptions, and conservation efforts. *Front. Psychol.* 10:1746
44. Colléony A, Clayton S, Couvet D, Saint Jalme M, Prévot AC. 2017. Human preferences for species conservation: Animal charisma trumps endangered status. *Biol. Conserv.* 206:263–69

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45. Pritchard DJ, Fa JE, Oldfield S, Harrop SR. 2012. Bring the captive closer to the wild: redefining the role of ex situ conservation. *Oryx* 46(1):18–23
46. Catibog-Sinha C. 2008. Zoo tourism: biodiversity conservation through tourism. *J. Ecotourism* 7(2–3):160–78
47. Conde DA, Flesness N, Colchero F, Jones OR, Scheuerlein A. 2011. An emerging role of zoos to conserve biodiversity. *Science* 331(6023):1390–91
48. Che-Castaldo JP, Grow SA, Faust LJ. 2018. Evaluating the contribution of North American zoos and aquariums to endangered species recovery. *Sci. Rep.* 8:9789
49. Barbanti A, Martin C, Blumenthal JM, Boyle J, Broderick AC, et al. 2019. How many came home? Evaluating ex situ conservation of green turtles in the Cayman Islands. *Mol. Ecol.* 28(7):1637–51
50. Kleiman DG. 1989. Reintroduction of captive mammals for conservation. *Bioscience* 39(3):152–61
51. Ebenhard T. 1995. Conservation breeding as a tool for saving animal species from extinction. *Trends Ecol. Evol.* 10(11):438–43
52. Conway WG. 2011. Buying time for wild animals with zoos. *Zoo Biol.* 30(1):1–8
53. Shier DM. 2015. Developing a standard for evaluating reintroduction success using IUCN Red List indices. *Anim. Conserv.* 18(5):411–12
54. Zimmermann A. 2010. The role of zoos in contributing to in situ conservation. In *Wild Mammals in Captivity: Principles and Techniques for Zoo Management*, ed. DG Kleiman, KV Thompson, CK Baer, pp. 281–87. Chicago: Univ. Chicago Press
55. Balmford A, Leader-Williams N, Green MJB. 1995. Parks or arks: Where to conserve threatened mammals? *Biodivers. Conserv.* 4(6):595–607
56. Robert A. 2009. Captive breeding genetics and reintroduction success. *Biol. Conserv.* 142(12):2915–22
57. Kelly JD. 1997. Effective conservation in the twenty-first century: the need to be more than a zoo. One organization's approach. *Int. Zoo Yearb.* 35(1):1–14
58. Santika T, Ancrenaz M, Wilson KA, Spehar S, Abram N, et al. 2017. First integrative trend analysis for a great ape species in Borneo. *Sci. Rep.* 7(1):4839
59. Armstrong DP, Seddon PJ. 2008. Directions in reintroduction biology. *Trends Ecol. Evol.* 23(1):20–25
60. Seddon PJ, Armstrong DP, Maloney RF. 2007. Developing the science of reintroduction biology. *Conserv. Biol.* 21(2):303–12
61. Braverman I. 2014. Conservation without nature: the trouble with *in situ* versus *ex situ* conservation. *Geoforum* 51:47–57
62. Int. Union Conserv. Nat. 2021. *The IUCN Red List of Threatened Species. Version 2021-3*. Fontainbleau, Fr.: Int. Union Conserv. Nat. **https://www.iucnredlist.org**
63. Tilman D, Clark M, Williams DR, Kimmel K, Polasky S, Packer C. 2017. Future threats to biodiversity and pathways to their prevention. *Nature* 546(7656):73–81
64. Assoc. Zoos Aquar. 2020. *Annual report on conservation and science: highlights*. Rep., Assoc. Zoos Aquar., Silver Spring, MD
65. Harding LE, Abu-Eid OF, Hamidan N, al Sha'lan A. 2007. Reintroduction of the Arabian oryx *Oryx leucoryx* in Jordan: war and redemption. *Oryx* 41(4):478–87
66. Parrott ML, Wicker LV, Lamont A, Banks C, Lang M, et al. 2021. Emergency response to Australia's Black Summer 2019–2020: the role of a zoo-based conservation organisation in wildlife triage, rescue, and resilience for the future. *Animals* 11(6):1515
67. Eur. Assoc. Zoos Aquar. 2020. *Annual report 2020*. Rep., Eur. Assoc. Zoos Aquar., Amsterdam
68. Gusset M, Dick G. 2010. "Building a future for wildlife"? Evaluating the contribution of the world zoo and aquarium community to *in situ* conservation. *Int. Zoo Yearb.* 44(1):183–91
69. MacDonald E. 2015. Quantifying the impact of Wellington Zoo's persuasive communication campaign on post-visit behavior. *Zoo Biol.* 34(2):163–69
70. Bowkett AE. 2009. Recent captive-breeding proposals and the return of the ark concept to global species conservation. *Conserv. Biol.* 23(3):773–76
71. Beck BB, Rapaport LG, Price MRS, Wilson AC. 1994. Reintroduction of captive-born animals. In *Creative Conservation: Interactive Management of Wild and Captive Animals*, ed. PJS Olney, GM Mace, ATC Feistner, pp. 265–86. London: Chapman & Hall

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72. Berger-Tal O, Saltz D. 2014. Using the movement patterns of reintroduced animals to improve reintroduction success. *Curr. Zool.* 60(4):515–26
73. Morell V. 2008. Conservation biology: into the wild: Reintroduced animals face daunting odds. *Science* 320(5877):742–43
74. Gilbert T, Soorae PS. 2017. Editorial: the role of zoos and aquariums in reintroductions and other conservation translocations. *Int. Zoo Yearb.* 51(1):9–14
75. Waugh DR. 2000. Parrot conservation and Loro Parque Fundacion, Puerto de la Cruz. *Int. Zoo Yearb.* 37(1):288–98
76. Resende PS, Viana-Junior AB, Young RJ, de Azevedo CS. 2020. A global review of animal translocation programs. *Anim. Biodivers. Conserv.* 43(2):221–32
77. Grant TD, Hudson RD. 2015. West Indian iguana *Cyclura* spp. reintroduction and recovery programmes: zoo support and involvement. *Int. Zoo Yearb.* 49(1):49–55
78. Moseby KE, Lollback GW, Lynch CE. 2018. Too much of a good thing; successful reintroduction leads to overpopulation in a threatened mammal. *Biol. Conserv.* 219:78–88
79. Ceballos G, Ehrlich PR, Dirzo R. 2017. Biological annihilation via the ongoing sixth mass extinction signaled by vertebrate population losses and declines. *PNAS* 114(30):E6089–96
80. Gusset M, Dick G. 2011. The global reach of zoos and aquariums in visitor numbers and conservation expenditures. *Zoo Biol.* 30(5):566–69
81. Packer J, Ballantyne R. 2010. The role of zoos and aquariums in education for a sustainable future. *New Dir. Adult Contin. Educ.* 2010(127):25–34
82. Patrick PG, Matthews CE, Ayers DF, Tunnicliffe SD. 2007. Conservation and education: prominent themes in zoo mission statements. *J. Environ. Educ.* 38(3):53–60
83. Roe K, McConney A. 2015. Do zoo visitors come to learn? An internationally comparative, mixed-methods study. *Environ. Educ. Res.* 21(6):865–84
84. Coe JC. 1985. Design and perception: making the zoo experience real. *Zoo Biol.* 4(2):197–208
85. Morgan JM, Hodgkinson M. 1999. The motivation and social orientation of visitors attending a contemporary zoological park. *Environ. Behav.* 31(2):227–39
86. Ballantyne R, Packer J. 2016. Visitors' perceptions of the conservation education role of zoos and aquariums: implications for the provision of learning experiences. *Visit. Stud.* 19(2):193–210
87. Whitehouse J, Waller BM, Chanvin M, Wallace EK, Schel AM, et al. 2014. Evaluation of public engagement activities to promote science in a zoo environment. *PLOS ONE* 9(11):e113395
88. Clayton S, Fraser J, Saunders CD. 2009. Zoo experiences: conversations, connections, and concern for animals. *Zoo Biol.* 28(5):377–97
89. Mellish S, Pearson EL, Sanders B, Litchfield CA. 2016. Marine wildlife entanglement and the Seal the Loop initiative: a comparison of two free-choice learning approaches on visitor knowledge, attitudes and conservation behaviour. *Int. Zoo Yearb.* 50(1):129–54
90. Andersen LL. 2003. Zoo education: from formal school programmes to exhibit design and interpretation. *Int. Zoo Yearb.* 38(1):75–81
91. Moss A, Esson M. 2010. Visitor interest in zoo animals and the implications for collection planning and zoo education programmes. *Zoo Biol.* 29(6):715–31
92. D'Cruze N, Khan S, Carder G, Megson D, Coulthard E, et al. 2019. A global review of animal-visitor interactions in modern zoos and aquariums and their implications for wild animal welfare. *Animals* 9(6):332
93. Fernandez EJ, Tamborski MA, Pickens SR, Timberlake W. 2009. Animal-visitor interactions in the modern zoo: conflicts and interventions. *Appl. Anim. Behav. Sci.* 120(1–2):1–8
94. Carr N. 2016. An analysis of zoo visitors' favourite and least favourite animals. *Tour. Manag. Perspect.* 20:70–76
95. Powell DM, Bullock EVW. 2015. Evaluation of factors affecting emotional responses in zoo visitors and the impact of emotion on conservation mindedness. *Anthrozoös* 27(3):389–405
96. Swanagan JS. 2010. Factors influencing zoo visitors' conservation attitudes and behavior. *J. Environ. Educ.* 31(4):26–31
97. Smith L, Broad S. 2008. Do zoo visitors attend to conservation messages? A case study of an elephant exhibit. *Tour. Rev. Int.* 11(3):225–35

Downloaded from www.annualreviews.org. Guest (guest) IP: 1.234.183.99 On: Mon, 03 Nov 2025 09:00:37

98. Davey G. 2006. Relationships between exhibit naturalism, animal visibility and visitor interest in a Chinese zoo. *Appl. Anim. Behav. Sci.* 96(1–2):93–102
99. Collins C, Corkery I, McKeown S, McSweeney L, Flannery K, et al. 2020. An educational intervention maximizes children's learning during a zoo or aquarium visit. *J. Environ. Educ.* 51(5):361–80
100. Miller LJ, Zeigler-Hill V, Mellen J, Koeppel J, Greer T, Kuczaj S. 2013. Dolphin shows and interaction programs: Benefits for conservation education? *Zoo Biol.* 32(1):45–53
101. Falk JH, Heimlich JE, Vernon CL, Bronnenkant K. 2010. Critique of a critique: Do zoos and aquariums promote attitude change in visitors? *Soc. Anim.* 18(4):415–19
102. Hacker CE, Miller LJ. 2016. Zoo visitor perceptions, attitudes, and conservation intent after viewing African elephants at the San Diego Zoo Safari Park. *Zoo Biol.* 35(4):355–61
103. Adelman LM, Falk JH, James S. 2000. Impact of National Aquarium in Baltimore on visitors' conservation attitudes, behavior, and knowledge. *Curator* 43(1):33–61
104. Lukas KE, Ross SR. 2005. Zoo visitor knowledge and attitudes toward gorillas and chimpanzees. *J. Environ. Educ.* 36(4):33–48
105. Moss A, Jensen E, Gusset M. 2015. Evaluating the contribution of zoos and aquariums to Aichi Biodiversity Target 1. *Conserv. Biol.* 29(2):537–44
106. Jensen E. 2014. Evaluating children's conservation biology learning at the zoo. *Conserv. Biol.* 28(4):1004–11
107. Learmonth MJ, Chiew SJ, Godinez A, Fernandez EJ. 2021. Animal-visitor interactions and the visitor experience: visitor behaviors, attitudes, perceptions, and learning in the modern zoo. *Anim. Behav. Cogn.* 8(4):632–49
108. Davidson SK, Passmore C, Anderson D. 2010. Learning on zoo field trips: the interaction of the agendas and practices of students, teachers, and zoo educators. *Sci. Educ.* 94(1):122–41
109. Dierking LD, Adelman LM, Ogden J, Lehnhardt K, Miller L, Mellen JD. 2004. Using a behavior change model to document the impact of visits to Disney's Animal Kingdom: a study investigating intended conservation action. *Curator* 47(3):322–43
110. Hughes K. 2013. Measuring the impact of viewing wildlife: Do positive intentions equate to long-term changes in conservation behaviour? *J. Sustain. Tour.* 21(1):42–59
111. Marino L, Lilienfeld SO, Malamud R, Nobis N, Broglio R. 2010. Do zoos and aquariums promote attitude change in visitors? A critical evaluation of the American Zoo and Aquarium Study. *Soc. Anim.* 18(2):126–38
112. Nygren NV, Ojalammi S. 2018. Conservation education in zoos: a literature review. *Trace* 4:62–76
113. Mallavarapu S, Taglialatela LA. 2019. A post-occupancy evaluation of the impact of exhibit changes on conservation knowledge, attitudes, and behavior of zoo visitors. *Environ. Educ. Res.* 25(10):1552–69
114. Stoinski TS, Allen MT, Bloomsmith MA, Forthman DL, Maple TL. 2010. Educating zoo visitors about complex environmental issues: Should we do it and how? *Curator* 45(2):129–43
115. Funk SM, Conde D, Lamoreux J, Fa JE. 2017. Meeting the Aichi targets: pushing for zero extinction conservation. *Ambio* 46(4):443–55
116. Kaufman AB, Bashaw MJ, Maple TL, eds. 2018. *Scientific Foundations of Zoos and Aquariums: Their Role in Conservation and Research*. Cambridge, UK: Cambridge Univ. Press
117. Rose PE, Brereton JE, Rowden LJ, de Figueiredo RL, Riley LM. 2019. What's new from the zoo? An analysis of ten years of zoo-themed research output. *Palgrave Commun*. 5:128
118. Welden HLÅ, Stelvig M, Nielsen CK, Purcell C, Eckley L, et al. 2020. The contributions of EAZA zoos and aquaria to peer-reviewed scientific research. *J. Zoo Aquar. Res.* 8(2):133–38
119. Loh T-L, Larson ER, David SR, de Souza LS, Gericke R, et al. 2018. Quantifying the contribution of zoos and aquariums to peer-reviewed scientific research. *Facets* 3(1):287–99
120. Finlay TW, Maple TL. 1986. A survey of research in American zoos and aquariums. *Zoo Biol.* 5(3):261–68
121. Anderson US, Maple TL, Bloomsmith MA. 2010. Factors facilitating research: a survey of zoo and aquarium professionals. *Zoo Biol.* 29(6):663–75
122. Anderson US, Kelling AS, Maple TL, Anderson U. 2008. Twenty-five years of *Zoo Biology*: a publication analysis. *Zoo Biol.* 27:444–57
123. Wemmer C, Rodden M, Pickett C. 1997. Publication trends in *Zoo Biology*: a brief analysis of the first 15 years. *Zoo Biol.* 16:3–8

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124. Hutchins M, Thompson SD. 2008. Zoo and aquarium research: priority setting for the coming decades. *Zoo Biol*. 27:488–97
125. Anzai W, Ban K, Hagiwara S, Kako T, Kashiwagi N, et al. 2022. Quantifying the 60-year contribution of Japanese zoos and aquariums to peer-reviewed scientific research. *Animals* 12(5):598
126. Ceballos G, Ehrlich PR, Raven PH. 2020. Vertebrates on the brink as indicators of biological annihilation and the sixth mass extinction. *PNAS* 117(24):13596–602
127. Titeux N, Brotons L, Settele J. 2019. IPBES promotes integration of multiple threats to biodiversity. *Trends Ecol. Evol.* 34(11):969–70
128. Burgass MJ, Arlidge WNS, Addison PFE. 2018. Overstating the value of the IUCN Red List for business decision-making. *Conserv. Lett.* 11(3):e12456
129. Pergl J, Brundu G, Harrower CA, Cardoso AC, Genovesi P, et al. 2020. Applying the Convention on Biological Diversity Pathway Classification to alien species in Europe. *NeoBiota* 62:333–63
130. Veasey JS. 2017. In pursuit of peak animal welfare; the need to prioritize the meaningful over the measurable. *Zoo Biol*. 36(6):413–25
131. Albert C, Luque GM, Courchamp F. 2018. The twenty most charismatic species. *PLOS ONE* 13(7):e0199149
132. Pino-del-Carpio A, Ariño AH, Villarroya A, Puig J, Miranda R. 2014. The biodiversity data knowledge gap: assessing information loss in the management of Biosphere Reserves. *Biol. Conserv.* 173:74–79

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